

Guiding Generative Probabilistic Weather Models

Alain Joss

SDSC

June 23, 2026

Introduction

Starting Question

Can we steer generative probabilistic weather models to generate extreme, physically realistic weather events?

① Steer

- Sampling-based guidance
- Guidance-based modification of the generation process

② Extreme

- Defined for a target region and variable(s)
- Evaluated relative to the forecast distribution

③ Physically Realistic

- Realism encouraged by the steering design
- Assessed using realism-related metrics

Goal: Scenario generation for downstream tasks.

Approach

① Steering

- ArchesWeather as the generative probabilistic weather model
- Guidance-based methods adapted from the image domain

② Extreme

- Mask defines the target region and forcing variable
- Change trajectory relative to the unguided forecast distribution
- Focus on heatwave scenarios

③ Physically Realistic

- Guidance is constrained or regularized
- Assessment through:
 - Visual checks
 - Distributional checks using power spectrum

Example

Guiding
Generative
Probabilistic
Weather
Models

Introduction

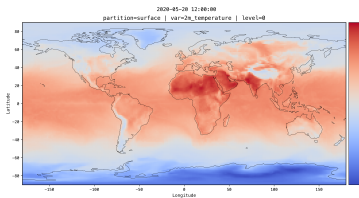
Background

Weather-
guidance design

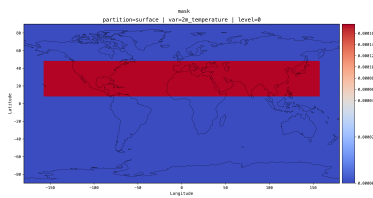
Guidance
methods

Experiments

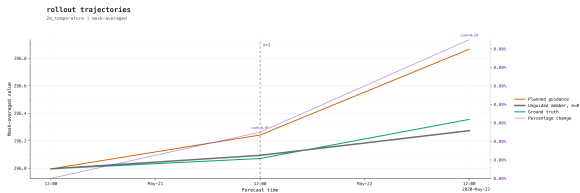
Results



Target variable



Target region



Guidance signal

Background

Model: ArchesWeather

Forecast decomposition

$$x_{n+1} = \mathbb{E}[x_{n+1} \mid x_n, x_{n-1}, n] + r_{n+1}$$

- Deterministic model predicts the expected next weather state.
- Residual flow model generates the stochastic residual r_{n+1} .
- Time information includes month and time of day.
- Spec:

Dataset	ERA5
Forecast step	24h
Grid	global 1.5° , 121×240 cells
Variables	4 surface, 6 pressure-level variables
Pressure levels	13 levels, 50–1000 hPa

Flow Matching

- Learned vector field $u_\theta(X_t, t)$ transports noise to data:

$$\frac{d}{dt}X_t = u_\theta(X_t, t)$$

- s.t. sampling generates residuals:

$$r_{n+1} \sim p_{\text{data}}$$

Algorithm 1 Euler Sampling

Require: Vector field u_θ , number of steps T

- 1: Sample $X_0 \sim p_{\text{init}}$
 - 2: $h \leftarrow 1/T$
 - 3: **for** $i = 0, \dots, T - 1$ **do**
 - 4: $t \leftarrow i/T$
 - 5: $X_{t+h} \leftarrow X_t + h u_\theta(X_t, t)$
 - 6: **end for**
 - 7: **return** X_1
-

Weather-guidance design

Guidance Design

- Guidance modifies the sampling process to favor samples with a desired property y .
- In our case, y corresponds to an extreme weather scenario:
 - target variable,
 - target region,
 - desired change over the unguided forecast.
- All guidance methods use this information to modify the vector field:

$$u_t(z_t) \longrightarrow \tilde{u}_t(z_t)$$

- The central design choice is therefore:

Question

How do we define the guidance objective y for weather trajectories?

Weather Guidance Objective

- We guide the generated state toward a prescribed change relative to the unguided forecast.
- The target is defined through a masked spatial average:

Weather Guidance Loss

$$\mathcal{L}_n^{\text{target}} = \left\| \frac{\sum_i m_i \hat{x}_{n,i}}{\sum_i m_i} - (1 + \delta_n) \frac{\sum_i m_i \hat{x}_{n,i}^{\text{ung}}}{\sum_i m_i} \right\|^2$$

- **m**: mask selecting target region and variable
- $\hat{x}_n$: current guided clean-state estimate
- $\hat{x}_n^{\text{ung}}$: unguided prediction
- δ_n : desired relative change at rollout step n

Loss evaluation

The loss is evaluated at different stages during guided sampling depending on the adopted method.

Loss Regularization

- The guidance loss can be regularized to discourage unrealistic deviations from the unguided forecast as follows:

$$\mathcal{L}_n^{\text{total}} = \mathcal{L}_n^{\text{target}} + \lambda \mathcal{R}$$

where $\mathcal{R}$ is the chosen regularization term.

Regularization Choices

$$\mathcal{R}_{\text{ID}} = \|\hat{x}_n - \hat{x}_n^{\text{ung}}\|^2$$

$$\mathcal{R}_{\text{PS}} = \|\text{PS}(\hat{x}_n) - \text{PS}(\hat{x}_n^{\text{ung}})\|^2$$

$$\mathcal{R}_{\text{A}} = \|\text{A}(\hat{x}_n) - \text{A}(\hat{x}_n^{\text{ung}})\|^2$$

- $\mathcal{R}_{\text{ID}}$: keeps the full guided state close to the unguided forecast.
- $\mathcal{R}_{\text{PS}}$: preserves the spatial power spectrum.
- $\mathcal{R}_{\text{A}}$: preserves activity-related structure.

Guiding Weather Rollouts

- Unlike image generation, weather generation evolves over multiple forecast steps.
- The desired change is therefore defined as a trajectory:

$$\delta_1, \delta_2, \dots, \delta_N$$

- At each rollout step n , the guidance objective becomes

$$\mathcal{L}_n = \mathcal{L}(\hat{x}_n, \hat{x}_n^{\text{ung}}, \delta_n, \mathbf{m})$$

- This allows us to prescribe how the extreme event should develop over time.
- We may play with different schemes to define $\{\delta\}_{n=1}^N$.

Notation summary

At rollout step n and flow time t :



- z_t : current flow state
- $\hat{x}_n$: clean estimate at rollout step n at flow time t
- $\mathcal{L}_n^{\text{total}}$: target loss plus optional regularization

Guidance Gradient

$$g_{n,t} = -\nabla_{z_t} \mathcal{L}_n^{\text{total}}(\hat{x}_n, y)$$

- All methods below use this gradient, but inject it into $\tilde{u}_t^\theta$ using different schemes.

Guidance methods

LBG: Loss-Based Guidance

Idea: modify the vector field using the gradient of a target likelihood.

$$p(x|y) \propto p(y|x)p(x)$$

$$\nabla_x \log p(x|y) = \nabla_x \log p(x) + \nabla_x \log p(y|x)$$

Guided Vector Field

$$\tilde{u}_t(z_t|y) = u_t(z_t) + w a_t \nabla_{z_t} \log p(y|z_t)$$

Using the weather loss:

$$\nabla_{z_t} \log p(y|z_t) = -\nabla_{z_t} \mathcal{L}_n(\hat{x}_n, y) = g_{n,t}$$

- For linear probability paths: $a_t = \frac{1-t}{t}$
- w controls guidance strength
- For $w \neq 1$, the exact probabilistic interpretation is lost
- Limitation: ignores uncertainty in $x_1|z_t$.

LBG-MC: Loss-Based Guidance Monte Carlo

Idea: avoid guiding only through the point estimate $\hat{x}_n$.

- The guidance likelihood should account for uncertainty in the clean state:

$$p(y|z_t) = \mathbb{E}_{\hat{x}_n|z_t} [\exp(-\mathcal{L}_n(\hat{x}_n, y))]$$

- Equivalently:

$$p(y|z_t) = \int p(\hat{x}_n|z_t) \exp(-\mathcal{L}_n(\hat{x}_n, y)) \, dx$$

- Since the true posterior $p(\hat{x}_n|z_t)$ is intractable, approximate it by

$$q(\hat{x}_n|z_t) = \mathcal{N}(\hat{x}_n, \sigma_t^2 I)$$

- In our implementation:

$$\sigma_t = R a_t$$

LBG-MC: Loss-Based Guidance Monte Carlo

Sample clean-state candidates:

$$x^{(i)} = \hat{x}_n + \sigma_t \varepsilon^{(i)}, \quad \varepsilon^{(i)} \sim \mathcal{N}(0, I), \quad \sigma_t = R a_t$$

Approximate the guidance likelihood:

$$p(y|z_t) \approx \frac{1}{N_{\text{MC}}} \sum_{i=1}^{N_{\text{MC}}} \exp\left(-\mathcal{L}_n(x^{(i)}, y)\right)$$

$$\nabla_{z_t} \log p(y|z_t) \approx \nabla_{z_t} \log \frac{1}{N_{\text{MC}}} \sum_{i=1}^{N_{\text{MC}}} \exp\left(-\mathcal{L}_n(x^{(i)}, y)\right)$$

Monte-Carlo Loss

$$\tilde{\mathcal{L}}_n = \log N_{\text{MC}} - \log \text{sumexp}_i \left(-\mathcal{L}_n(x^{(i)}, y) \right)$$

- N_{MC} : number of Monte-Carlo samples
- R : spread magnitude; a_t shrinks the cloud over the flow
- Limitation: Gaussian posterior approximation

UG: Universal Guidance

Idea: optimize noisy state before taking the generative step.

Inner Optimization

$$\Delta z_t^* = \arg \min_{\Delta z} \mathcal{L}_n(\hat{x}_{n,t}(z_t + \Delta z), y)$$

In practice, approximate Δz_t^* with K gradient steps:

$$\Delta z_t \leftarrow \Delta z_t - \eta \nabla_{\Delta z} \mathcal{L}_n(\hat{x}_{n,t}(z_t + \Delta z), y)$$

Then apply the optimized correction:

$$z_t^{\text{gui}} = z_t + w a_t \Delta z_t$$

Strong correction may leave the data manifold $\rightarrow$ re-inject noise.

Bansal et al., Universal Guidance for Diffusion Models, CVPR Workshop 2023.

UG: Universal Guidance

Idea: repeat guidance at the same flow time t before moving to $t + h$, in a self recurrent fashion.

One Self-Recurrence Step

$$z_t^{(s)} \xrightarrow{\text{guide}} z_t^{(s),\text{gui}} = z_t^{(s)} + w a_t \Delta z_t^{(s)}$$

$$z_t^{(s),\text{gui}} \xrightarrow{\text{Euler}} z_{t+h}^{(s)} = z_t^{(s),\text{gui}} + h u_t(z_t^{(s),\text{gui}})$$

$$z_{t+h}^{(s)} \xrightarrow{\text{renoise}} z_t^{(s+1)}$$

- Repeat this loop for $s = 1, \dots, S$ at the same flow time t .
- Re-noising brings the sample back to the same flow level after a "trial" step.
- After the last recurrence, keep the Euler step and move to $t + h$.
- Hyperparameters: K inner optimization steps, S recurrence steps, guidance strength w , learning rate η

Universal Guidance: Algorithm

Algorithm 2 Universal Guidance with Self-Recurrence

Require: Vector field u_θ , loss $\mathcal{L}_n$, clean predictor $\hat{x}_{n,t}$, steps T , inner steps K , recurrences S

```
1: Sample  $z_0 \sim p_{\text{init}}$ 
2:  $h \leftarrow 1/T$ 
3: for  $j = 0, \dots, T - 1$  do
4:    $t \leftarrow j/T$ 
5:   for  $s = 1, \dots, S$  do
6:      $\Delta z_t \leftarrow 0$ 
7:     for  $k = 1, \dots, K$  do
8:        $\Delta z_t \leftarrow \Delta z_t - \eta \nabla_{\Delta z_t} \mathcal{L}_n(\hat{x}_{n,t}(z_t + \Delta z_t), y)$ 
9:     end for
10:     $z_t^{\text{gui}} \leftarrow z_t + w a_t \Delta z_t$ 
11:     $z_{t+h} \leftarrow z_t^{\text{gui}} + h u_\theta(z_t^{\text{gui}}, t)$ 
12:    if  $s < S$  then
13:       $z_t \leftarrow \text{renoise}(z_{t+h}, t + h \rightarrow t)$ 
14:    end if
15:  end for
```

FlowGrad (FG)

Guiding
Generative
Probabilistic
Weather
Models

Introduction

Background

Weather-
guidance design

**Guidance
methods**

Experiments

Results

FlowGrad-free (FGF)

Guiding
Generative
Probabilistic
Weather
Models

Introduction

Background

Weather-
guidance design

**Guidance
methods**

Experiments

Results

Methods Overview

Method	Mechanism	Cost / flow step
LBG	loss gradient at $\hat{x}_n$	1 backward
LBG-MC	MC-averaged likelihood gradient	S backward
UG	inner-loop latent shift Δz_t	$K \times S$ backward
FG	optimized schedule λ_t on $g_{n,t}$	backprop through ODE
FGF	free controls c_t	backprop through ODE

- **Score-based:** LBG and LBG-MC add a guidance gradient to the vector field.
- **Optimization-based:** UG, FG, and FGF optimize correction variables during sampling.

Method-Specific Hyperparameters

Guiding
Generative
Probabilistic
Weather
Models

Introduction

Background

Weather-
guidance design

Guidance
methods

Experiments

Results

Symbol	Meaning	Used in
w	guidance strength	LBG, LBG-MC, UG
a_t	flow-time guidance scaling	LBG, LBG-MC, UG
N_{MC}	Monte-Carlo samples	LBG-MC
R	posterior spread magnitude	LBG-MC
K	inner optimization steps	UG
S	self-recurrence steps	UG
η	learning rate	UG

Experiments

Experimental Setup

Parameter	Meaning
Start time	input-state timestamp at $n = 0$
M	number of ensemble members
N	number of rollout steps
T	number of flow steps per rollout step
Reference	unguided forecast
$\{\delta_n\}_{n=1}^N$	prescribed target change trajectory
λ	regularization strength
$\mathcal{R}$	regularization type
Mask	target region and variable
Mask mode	BBOX or Gaussian mask

Hyperparameter tuning

Guiding
Generative
Probabilistic
Weather
Models

Introduction

Background

Weather-
guidance design

Guidance
methods

Experiments

Results

Results

- Guiding Generative Probabilistic Weather Models
- Introduction
- Background
- Weather-guidance design
- Guidance methods
- Experiments
- Results**

Results

Physical Realism

Guiding
Generative
Probabilistic
Weather
Models

Introduction

Background

Weather-
guidance design

Guidance
methods

Experiments

Results

Conclusion

Guiding
Generative
Probabilistic
Weather
Models

Introduction

Background

Weather-
guidance design

Guidance
methods

Experiments

Results

QA

Appendix

Background

Flow Matching

- Assume observations are samples from a data distribution

$$p_{\text{data}}$$

- Generation corresponds to sampling from this distribution
- Start from a simple distribution

$$p_{\text{init}}$$

- Learn a transformation

$$p_{\text{init}} \longrightarrow p_{\text{data}}$$

Representing the Transformation

- The transformation is represented by a **vector field**

$$(t, x) \mapsto u_t(x)$$

- The vector field defines a trajectory

$$t \mapsto x_t$$

- Through the ODE

$$\frac{d}{dt}x_t = u_t(x_t)$$

- Solving the ODE induces a flow

$$x_t = \phi_t(x_0)$$

Sampling

- Learn a neural vector field

$$u_{\theta}(x_t, t)$$

- Solving

$$\frac{d}{dt}x_t = u_{\theta}(x_t, t)$$

transports samples from

$$p_{\text{init}} \rightarrow p_{\text{data}}$$

- The solution is generally not available in closed form.
- Numerical integration is therefore required.

Euler Integration

Algorithm 3 Sampling from a Flow Model

Require: Vector field u_θ , number of steps T

- 1: Sample $X_0 \sim p_{\text{init}}$
 - 2: $h \leftarrow 1/T$
 - 3: **for** $i = 0, \dots, T - 1$ **do**
 - 4: $t \leftarrow i/T$
 - 5: $X_{t+h} \leftarrow X_t + h u_\theta(X_t, t)$
 - 6: **end for**
 - 7: **return** X_1
-

From Sampling to Training

- During sampling, the learned ODE transports

$$X_0 \sim p_{\text{init}} \longrightarrow X_1 \sim p_{\text{data}}$$

- To train the vector field, we prescribe how intermediate samples should look:

$$p_0 = p_{\text{init}} \longrightarrow p_t \longrightarrow p_1 = p_{\text{data}}$$

- This family of intermediate distributions is the **probability path**.
- Flow matching trains u_θ to match the velocity of this path.

Training Flow Matching

- Define a Gaussian probability path conditioned on a sample x_1 :

$$p_t(x \mid x_1) = \mathcal{N}(x; tx_1, (1-t)^2 I)$$

- Equivalently, sample along this path using

$$x_t = (1-t)x_0 + tx_1, \quad x_0 \sim \mathcal{N}(0, I), \quad x_1 \sim p_{\text{data}}$$

- The target velocity along this path is

$$u_t^*(x_t) = \frac{d}{dt}x_t = x_1 - x_0$$

- Train u_θ to match this velocity:

$$\mathcal{L}_{FM}(\theta) = \mathbb{E}_{t, x_0, x_1} \left[\|u_\theta(x_t, t) - (x_1 - x_0)\|^2 \right]$$

From Loss to Guidance Gradient

- Once the target and regularization terms are chosen, the total loss defines the guidance signal:

$$\mathcal{L}_n^{\text{total}} = \mathcal{L}_n^{\text{target}} + \lambda \mathcal{R}$$

- We define the guidance likelihood through this loss:

$$p(y|z_t) = \exp(-\mathcal{L}_n^{\text{total}}(\hat{x}_{n,t}, y))$$

- Therefore, the guidance gradient is:

$$\nabla_{z_t} \log p(y|z_t) = -\nabla_{z_t} \mathcal{L}_n^{\text{total}}(\hat{x}_{n,t}, y)$$

Guidance Signal

$$g_{n,t} = -\nabla_{z_t} \mathcal{L}_n^{\text{total}}(\hat{x}_{n,t}, y)$$

LBG: Loss-Based Guidance

Idea: same loss-gradient guidance as classifier guidance, but with a normalized guidance scale.

LBG Update

$$\tilde{u}_t(z_t) = u_t(z_t) + \psi_t g_{n,t}, \quad g_{n,t} = -\nabla_{z_t} \mathcal{L}_n(\hat{x}_{n,t}, y)$$
$$\psi_t = \frac{w a_t}{\|g_{n,t}\| + \varepsilon}$$

- Hyperparameters: guidance strength w .
- $\hat{x}_{n,t}$ is treated as a point estimate of the clean state.
- Normalization stabilizes guidance magnitude across flow steps.
- Limitation: ignores uncertainty in $x_1|z_t$.

Chung et al., Diffusion Posterior Sampling for General Noisy Inverse Problems, ICLR 2023.